Inner products

On $\mathbb{R}^n$ $\left\langle \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \right\rangle = u_1 v_1 + \cdots + u_n v_n$

$$= \sum_{j=1}^{n} u_j v_j$$

On $\mathbb{C}^n$ $\left\langle \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \right\rangle = u_1 \overline{v_1} + \cdots + u_n \overline{v_n} = \sum_{j=1}^{n} u_j \overline{v_j}$

$0 \leq \|u\| = \sqrt{\langle u, u \rangle}$ = <u>norm</u> of u = <u>length</u> of u

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Def. Suppose $v \in \mathbb{R}^n$ or $\mathbb{C}^n$ is non-zero. Then we <u>normalize</u> v to obtain

$$\frac{1}{\|v\|} v$$

Recall that a normalized vector has length 1 (<u>unit vector</u>)

since $\left\| \frac{1}{\|v\|} v \right\| = \left| \frac{1}{\|v\|} \right| \|v\| = \frac{1}{\cancel{\|v\|}} \cancel{\|v\|} = 1.$

$\|v\| > 0$

Example

Take $v = \begin{bmatrix} 1 \\ 1+2i \end{bmatrix} \in \mathbb{C}^2$ (non zero)

$$\|v\| = \sqrt{\langle v, v \rangle} = \sqrt{(1)(\overline{1}) + (1+2i)\overline{(1+2i)}}$$

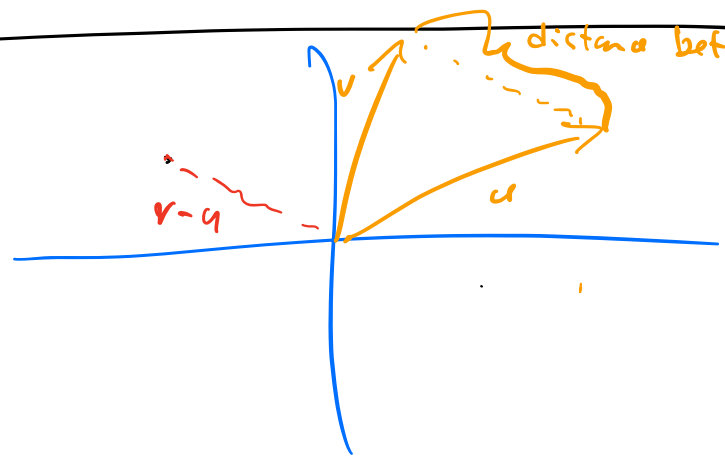
$$= \sqrt{1 + (1+2i)(1-2i)}$$

$$= \sqrt{1 + 1^2 + 2^2}$$

$$= \sqrt{6}.$$

To normalize v we get

$$\frac{1}{\|v\|} v = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1+2i \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} + 2/\sqrt{6}\, i \end{bmatrix}$$



distance between two vectors in $\mathbb{R}^2$.

$= \|v-u\|$

$= \|u-v\|$

Def. The <u>distance</u> between vectors u and v in $\mathbb{R}^n$ or $\mathbb{C}^n$ is

$$\operatorname{dist}(u,v) = \|u-v\|$$

$$= \sqrt{\langle u-v, u-v \rangle}$$

Remark

$$\|u-v\| = \|(-1)(-u+v)\|$$

$$= |-1| \, \|-u+v\|$$

$$= (1) \, \|v-u\|$$

$$= \|v-u\|$$

Example

$$\operatorname{dist}\left(\begin{bmatrix}1\\-2\end{bmatrix}, \begin{bmatrix}1\\1+i\end{bmatrix}\right) = \left\| \begin{bmatrix}1\\-2\end{bmatrix} - \begin{bmatrix}1\\1+i\end{bmatrix} \right\|$$

$$= \left\| \begin{bmatrix}0\\3-i\end{bmatrix} \right\|$$

$$= \sqrt{(0)\overline{(0)} + (3-i)\overline{(3-i)}}$$

$$= \sqrt{0 + (3-i)(3+i)}$$

$$= \sqrt{3^2 + 1^2}$$

$$= \sqrt{10}.$$

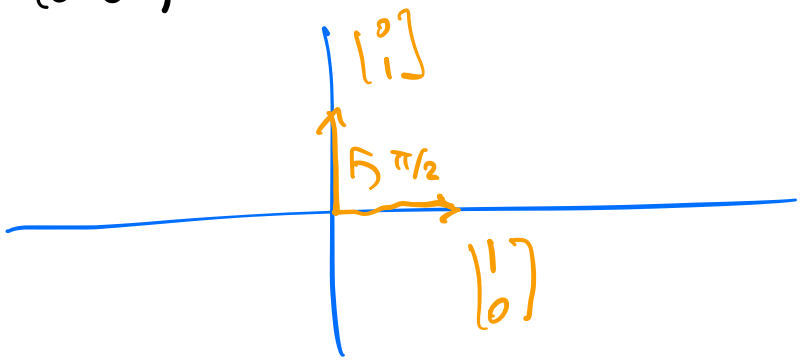
Recall from $\mathbb{R}^n$ that two vectors $u, v \in \mathbb{R}^n$ are <u>orthogonal</u> if $\langle u, v \rangle = 0$.

Def. Suppose $u, v \in \mathbb{R}^n$ or $\mathbb{C}^n$ then u and v are <u>orthogonal</u> if $\langle u, v \rangle = 0$.

Example

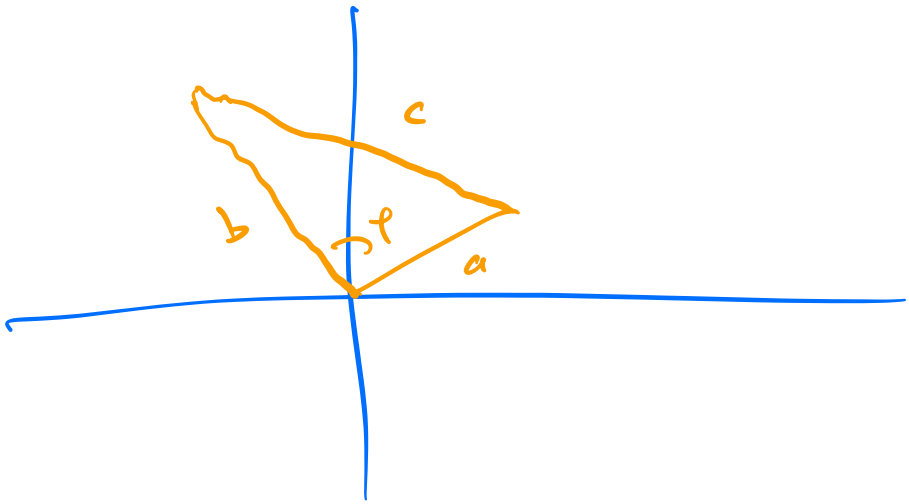
$$\left\langle \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix} \right\rangle = (1)(0) + (0)(1) = 0$$

$\Rightarrow$ $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$ are orthogonal.

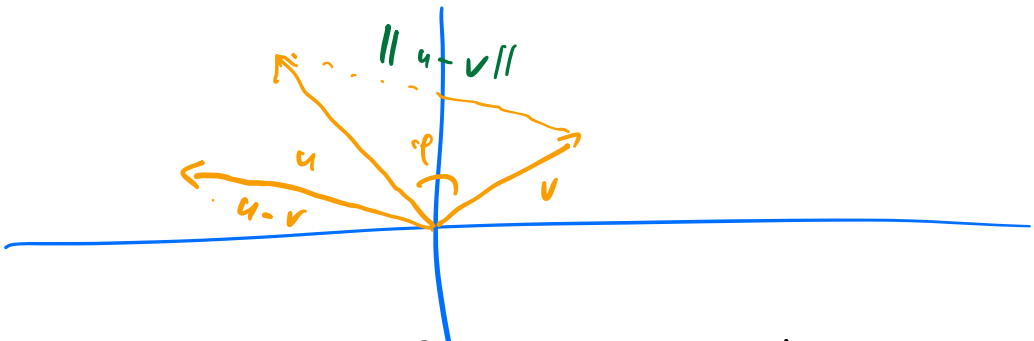


Angles and real inner products

Let's take for granted the cosine law.



$$c^2 = a^2 + b^2 - 2ab\cos\varphi.$$



$$\|u-v\|^2 = \|v\|^2 + \|u\|^2 - 2\|v\|\|u\| \cos\varphi.$$

$$\langle u-v, u-v\rangle = \langle v,v\rangle + \langle u,u\rangle - 2\|v\|\|u\|\cos\varphi$$

$$\cancel{\langle u,u\rangle} - \langle v,u\rangle - \langle u,v\rangle + \cancel{\langle v,v\rangle} = \cancel{\langle v,v\rangle} + \cancel{\langle u,u\rangle} - 2\|u\|\|v\|\cos\varphi$$

$$\cancel{-2}\langle u,v\rangle = \cancel{-2}\|u\|\|v\|\cos\varphi$$

$$\langle u,v\rangle = \|u\|\,\|v\|\cos\varphi.$$

$$\varphi = \arccos\left(\frac{\langle u,v\rangle}{\|u\|\,\|v\|}\right) \quad (\#)$$

What about two vectors in $\mathbb{R}^n$. In that case the two vector lie in a subspace $W = \text{span}\{u,v\}$ of dimension 2. $W \cong \mathbb{R}^2$ (coordinate maps) We can convert the picture in W into the picture for $\mathbb{R}^2$ above using the coordinate map.

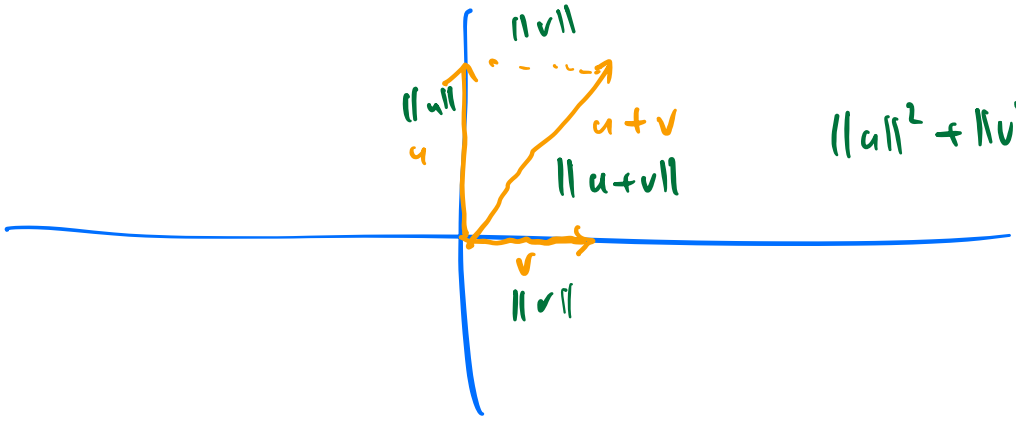
Alternatively, we can <u>define</u> the angle between to vectors in $\mathbb{R}^n$ by (#) above.

For complex vectors the concept of angle is not useful. However, the concept of orthogonality ($\langle u,v\rangle = 0$) is still useful.

Theorem 2 [Pythagorean Thm]

(a) $u, v \in \mathbb{R}^n$ are orthogonal $\iff \|u\|^2 + \|v\|^2 = \|u+v\|^2$

(b) If $u, v \in \mathbb{C}^n$ are orthogonal then $\|u\|^2 + \|v\|^2 = \|u+v\|^2$



$\|u\|^2 + \|v\|^2 = \|u+v\|^2.$

Sketch of proof of (a)

Suppose $u, v \in \mathbb{R}^n$ are orthogonal i.e. $\langle u, v \rangle = 0$

Then
$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v \rangle \\ &= \langle u, u \rangle + \cancel{\langle v, u \rangle}^{0} + \cancel{\langle u, v \rangle}^{0} + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2. \end{aligned}$$

Def. Suppose W is a subspace of $\mathbb{R}^n$ or $\mathbb{C}^n$.

Then the <u>orthogonal complement</u> of W

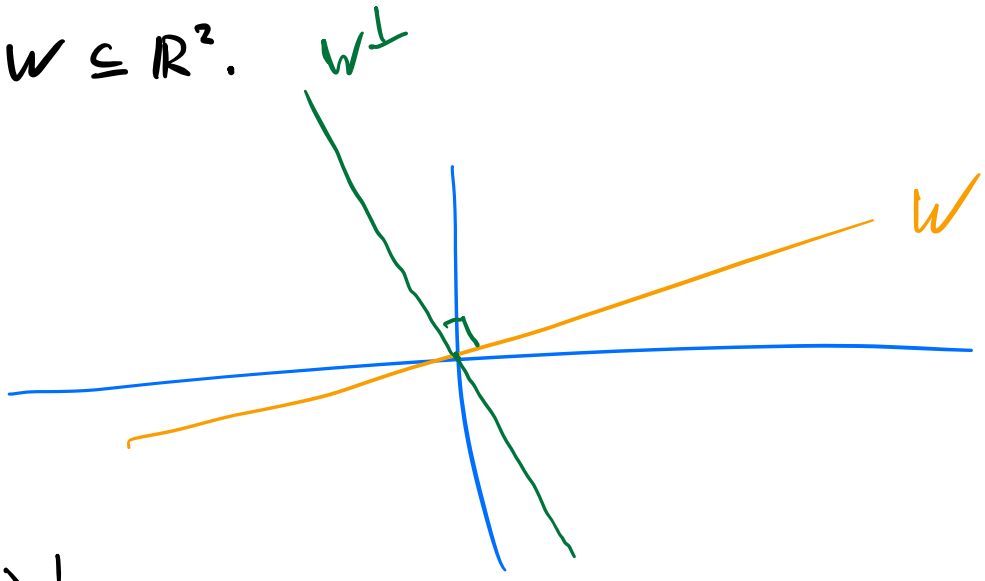
is $W^\perp = \{ v \in \mathbb{R}^n \text{ (or } \mathbb{C}^n) : \langle v, w \rangle = 0 \text{ for all } w \in W \}.$

"W perp"

all vectors which are orthogonal to every vector in W.

Example

Take $W \subseteq \mathbb{R}^2$.

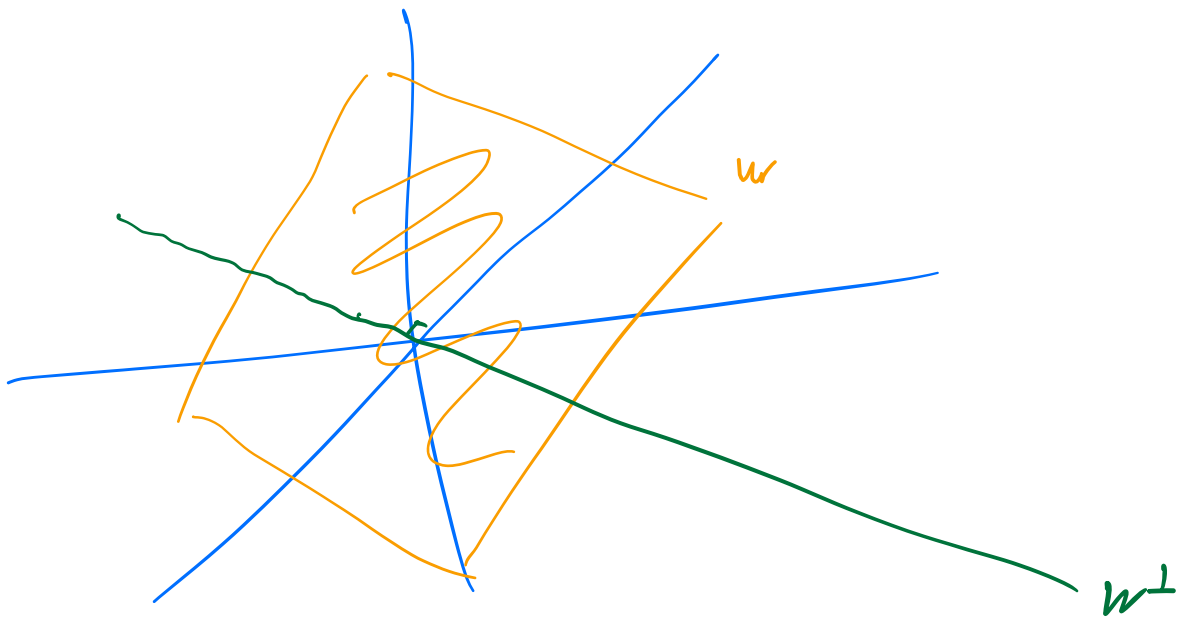


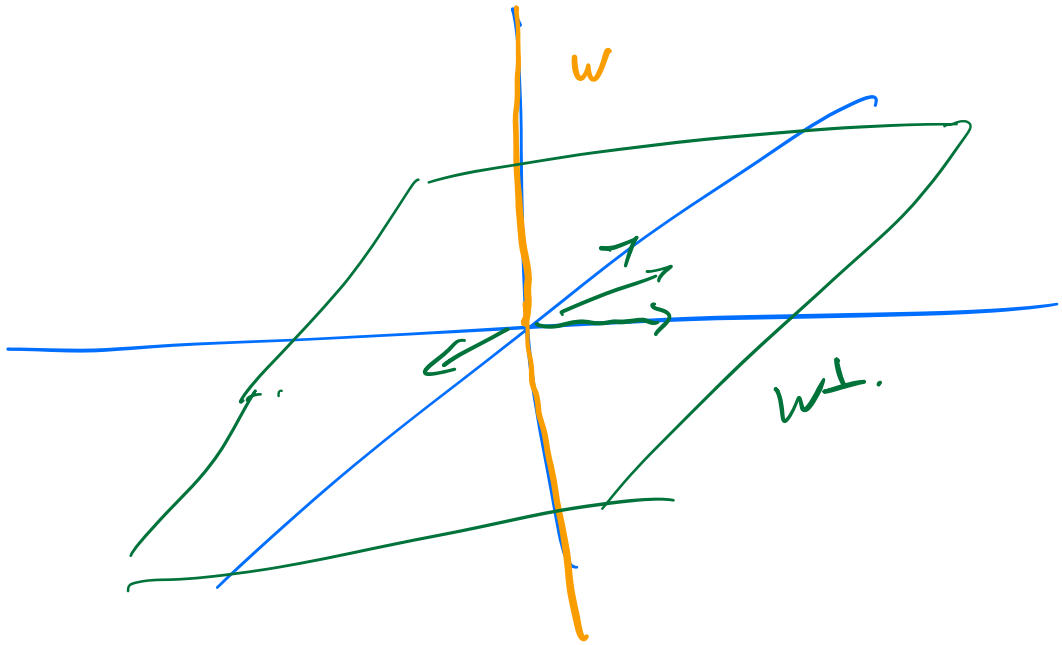
$$(\mathbb{R}^2)^\perp = \left\{\begin{bmatrix}0\\0\end{bmatrix}\right\}$$

$$\left\{\begin{bmatrix}0\\0\end{bmatrix}\right\}^\perp = \mathbb{R}^2$$

(Exercises)

Example





Exercises

① Suppose $W = \text{Span}\{w_1, \dots, w_k\} \subseteq \mathbb{C}^n$ or $\mathbb{R}^n$

Then $v \in W^\perp \iff \langle v, w_1 \rangle = \dots = \langle v, w_k \rangle = 0$.

② $W^\perp$ is a subspace of $\mathbb{R}^n$ or $\mathbb{C}^n$.

Example

Let $W = \text{Span}\left\{ \begin{bmatrix} 1 \\ i \end{bmatrix} \right\} \subseteq \mathbb{C}^2$.

By exercise ① above $v \in W^\perp \iff \left\langle v, \begin{bmatrix} 1 \\ i \end{bmatrix} \right\rangle = 0$

Let's find when $\left\langle v, \begin{bmatrix} 1 \\ i \end{bmatrix} \right\rangle = 0$.

Let $v = \begin{bmatrix} a_1 + b_1 i \\ a_2 + b_2 i \end{bmatrix} \in \mathbb{C}^2$.

Let's compute

$$\left\langle v, \begin{bmatrix} 1 \\ i \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} a_1+b_1 i \\ a_2+b_2 i \end{bmatrix}, \begin{bmatrix} 1 \\ i \end{bmatrix} \right\rangle$$

$$= (a_1+b_1 i)\overline{(1)} + (a_2+b_2 i)\overline{(i)}$$

$$= (a_1+b_1 i) + (a_2+b_2 i)(-i)$$

$$= a_1+b_1 i - a_2 i + b_2$$

$$= (a_1+b_2) + (b_1-a_2)i$$

So $0 = \left\langle v, \begin{bmatrix} 1 \\ i \end{bmatrix} \right\rangle \iff 0 = (a_1+b_2) + (b_1-a_2)i$

$$\iff \begin{cases} 0 = a_1+b_2 \\ 0 = b_1-a_2 \end{cases}$$

$$\iff \begin{cases} b_2 = -a_1 \\ a_2 = b_1 \end{cases}$$

$$\iff v = \begin{bmatrix} a_1+b_1 i \\ b_1-a_1 i \end{bmatrix} = \begin{bmatrix} a_1+b_1 i \\ (a_1+b_1 i)(-i) \end{bmatrix}$$

$$\iff v = (a_1+b_1 i)\begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\iff v \in z\begin{bmatrix} 1 \\ -i \end{bmatrix}, \quad z = a_1+b_1 i \in \mathbb{C}$$

$$\iff v \in \operatorname{span}\left\{\begin{bmatrix} 1 \\ -i \end{bmatrix}\right\}$$

Therefore $W^{\perp} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ -i \end{bmatrix}\right\} \subseteq \mathbb{C}^2$.